IoT & AI-BASED SMART AGRICULTURE PLATFORM FOR YIELD, DISEASE, IRRIGATION AND PEST MANAGEMENT

INDIVIDUAL THESIS

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Individual component: Disease Detection & Risk Forecasting

Group ID: R26-IT-157

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DECLARATION AND SUPERVISOR CERTIFICATION

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Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

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Co-supervisor: [Name]		

ABSTRACT

Paddy is Sri Lanka's staple crop, yet rice blast and sheath blight cause yield losses of 20-40% during humid monsoon periods, and most farm-level detection still depends on visual scouting after symptoms have already spread. Existing mobile diagnosis tools are trained mostly on global datasets and identify disease only after damage appears, while weather advisories do not translate environmental stress into disease probability. This thesis reports the individual component Disease Detection & Risk Forecasting within an integrated IoT and AI smart agriculture platform for yield, disease, irrigation and pest management. The aim is to provide paddy farmers with seven- to fourteen-day outbreak warnings by combining smartphone leaf-image classification with environmental risk forecasting.

A design-science approach was followed. A ResNet-50 convolutional neural network, trained by transfer learning on 224 x 224 field images, classifies leaves into four classes (healthy, rice blast, sheath blight and bacterial leaf disease). A Long Short-Term Memory (LSTM) network then forecasts outbreak probability from 14-day sequences of humidity, temperature, rainfall and recent CNN severity signals, so that visible symptoms and environmental triggers reinforce each other. Outputs are exposed through shared JSON APIs to the yield, irrigation and pest modules and are delivered to farmers through a Sinhala and English mobile interface with confidence levels and treatment guidance. Targets are an F1-score of at least 0.90 for classification, an AUC of at least 0.85 for outbreak prediction, a response time under three seconds, and farmer satisfaction above 80% in a Yala 2026 pilot.

[Insert principal quantitative results here once final evaluation is complete: F1-score, AUC, latency and pilot satisfaction.] The component's individual contribution is a locally adapted, fused image-and-weather forecasting pipeline that moves disease management from reactive identification to preventive, evidence-based action.

Keywords: rice blast, sheath blight, ResNet-50, LSTM, disease risk forecasting, smart agriculture

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My appreciation extends to the paddy farmers, agricultural extension officers and the Department of Agriculture and meteorological institutions whose knowledge, field access and validation feedback supported the study. [State any funding or substantial technical assistance received, or delete this sentence.]

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the (ROC) Curve
CNN	Convolutional Neural Network
DOA	Department of Agriculture (Sri Lanka)
FR	Functional Requirement
GRU	Gated Recurrent Unit
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
IoT	Internet of Things
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
NFR	Non-Functional Requirement
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
SO	Specific Objective
UI	User Interface
WBS	Work Breakdown Structure

1 INTRODUCTION

1.1 Overall project context

Rice is the staple food of Sri Lanka, and paddy cultivation supports a large population of smallholder farmers across the Yala and Maha seasons. Production is exposed to climate variability, disease, pest pressure and inefficient water use, while extension services cannot continuously monitor every field [5]. The group project (Project Code R26-IT-157) responds to these challenges with an IoT and AI-based smart agriculture platform for yield, disease, irrigation and pest management.

The platform is organised into four layers: a field and data capture layer (farmer smartphones and IoT sensors for humidity, temperature and rainfall), a data and integration layer (upload API, sensor gateway, metadata tagging and a PostgreSQL time-series store), an AI and analytics layer, and a delivery and user layer (a Sinhala and English farmer application, extension dashboards and API outputs). The AI and analytics layer contains four components: yield forecasting, disease detection and risk forecasting, irrigation scheduling, and pest management. The components exchange information through shared APIs so that, for example, a forecast disease outbreak can influence spray-window-aware irrigation and expected-yield estimates.

1.2 Individual component context

This thesis concerns the Disease Detection & Risk Forecasting component, which is the disease-intelligence service of the platform. It receives farmer leaf photographs together with IoT sensor streams and external meteorological data. It outputs a disease class, a severity estimate, and seven- and fourteen-day outbreak probabilities that are consumed by the yield, irrigation and pest-management components and presented to farmers and extension officers.

The component targets the diseases that most affect Sri Lankan paddy under humid, wet-canopy conditions, namely rice blast and sheath blight, together with bacterial leaf disease, and distinguishes them from healthy leaves. Its distinctive feature is that image-based evidence and environmental history are modelled jointly rather than treated as separate sources of information.

1.3 Problem statement

Rice blast and sheath blight cause yield losses of 20-40% during humid monsoon periods, and they spread quickly under wet-canopy conditions. Smallholder farmers usually recognise infection visually only after a meaningful portion of the leaf surface has already been damaged, which narrows the window for preventive intervention. Because extension officers cannot scout every field, late confirmation commonly leads to chemical overuse, higher production cost and avoidable yield loss.

The problem addressed in this thesis is therefore the absence of an accessible, early-warning capability that tells a paddy farmer, one to two weeks in advance, how likely a damaging outbreak is in his or her field, and what to do about it.

1.4 Research gap

Prior work shows that CNN architectures reach high plant-disease classification accuracy on curated images [1], [2], but field performance usually declines because of lighting variation, leaf angle, background clutter and mixed symptom severity. Time-series models such as LSTM have shown promise for environmental risk forecasting [6], yet disease-specific implementations for Sri Lankan paddy fields are limited and rarely integrated with mobile decision support. Commercial mobile applications such as Plantix provide usable diagnosis at scale, but they are post-symptomatic and mostly based on global datasets. Weather-only advisories, in turn, do not translate environmental stress into disease probability.

No readily deployable system currently combines smartphone leaf-image classification with Sri Lanka-specific environmental risk forecasting to generate seven- to fourteen-day outbreak warnings for paddy farmers. This component addresses that gap through integrated modelling, localised alert thresholds and interoperability with the wider platform.

1.5 Aim and objectives

Aim. To develop a Disease Detection & Risk Forecasting module that gives Sri Lankan paddy farmers early outbreak warnings one to two weeks ahead, using CNN-based image classification and LSTM-based environmental risk modelling.

The specific objectives and their measurable success indicators are given in Table 1.1.

Table 1.1: Specific objectives and success indicators

ID	Specific objective	Success indicator
SO1	Collect rice leaf images and environmental data for training and evaluation.	Dataset of 5,000+ field images and synchronised sensor records
SO2	Train a 4-class CNN for healthy, blast, sheath blight and bacterial-leaf-disease detection.	F1-score >= 0.90 on held-out evaluation
SO3	Develop an LSTM risk forecasting model for outbreak probability estimation.	AUC >= 0.85 for outbreak prediction
SO4	Integrate disease outputs into the platform APIs and farmer application.	Live Sinhala / English alerts and JSON API outputs
SO5	Validate field usefulness during the Yala 2026 season.	Farmer satisfaction above 80% in pilot feedback

1.6 Research questions or hypotheses

The study is guided by the following research questions:

1. RQ1: Can a ResNet-50 model trained on field images of Sri Lankan paddy classify healthy, blast, sheath blight and bacterial leaf disease with an F1-score of at least 0.90 on unseen farms?
2. RQ2: Can an LSTM trained on 14-day sequences of humidity, temperature, rainfall and CNN severity signals predict outbreaks seven to fourteen days ahead with an AUC of at least 0.85?
3. RQ3: Does feeding CNN severity outputs into the LSTM improve forecasting performance compared with using environmental variables alone?
4. RQ4: Do farmers find the Sinhala / English alert workflow useful and understandable in a real Yala 2026 pilot?

Correspondingly, the study tests two hypotheses. H1: the fused image-plus-weather model achieves a higher AUC than a weather-only LSTM baseline. H2: the CNN meets the F1-score target of 0.90 on a held-out field test set from unseen farms.

1.7 Scope, limitations and assumptions

In scope: four-class leaf classification (healthy, blast, sheath blight, bacterial leaf disease); seven- and fourteen-day outbreak probability forecasting; High, Medium and Low alert levels; API outputs to the yield, irrigation and pest modules; a Sinhala / English interface; and a pilot validation in Yala 2026.

Out of scope: diseases other than the four classes, pest identification (handled by the pest-management component), automated pesticide dosing and crops other than paddy.

Assumptions and limitations: farmers have access to a smartphone and intermittent connectivity; sensor nodes and meteorological feeds provide time-stamped humidity, temperature and rainfall data; outbreak labels are derived from field observation and agronomist confirmation; and results from a single-season pilot may not generalise to all agro-ecological zones.

1.8 Individual contribution

My individual contribution to the group project consists of: (a) assembling and curating the rice leaf image and environmental dataset; (b) designing and training the ResNet-50 disease classifier with augmentation tuned to field conditions; (c) designing the LSTM outbreak-risk model and the ensemble logic that reuses CNN severity as a temporal feature; (d) defining alert thresholds and Sinhala / English alert content; (e) building the shared JSON API through which other components consume disease outputs; and (f) planning and conducting the Yala 2026 field validation.

1.9 Thesis structure

Chapter 2 reviews the literature and existing systems. Chapter 3 describes the research methodology. Chapter 4 presents the requirements and solution design. Chapter 5 covers implementation, integration and testing. Chapter 6 reports results and validation. Chapter 7 discusses the findings, contributions and limitations. Chapter 8 presents commercialisation and sustainability considerations, and Chapter 9 concludes the thesis and outlines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review covers three areas: deep-learning image classification for plant and rice disease, time-series modelling of environmental disease risk, and mobile decision-support tools for farmers. Sources were drawn from journals and repositories in computer vision, remote sensing and agricultural informatics, and from national agricultural statistics. Studies were included if they addressed rice or plant-disease recognition, sequence-based forecasting of agro-meteorological risk, or field deployment. Studies focused only on laboratory imagery were retained as baselines but weighted lower for field relevance. [Extend with the full search strategy, databases, keywords, date range and inclusion/exclusion counts used.]

2.2 Domain and theoretical foundations

2.2.1 Major paddy diseases

Rice blast, caused by a fungal pathogen, and sheath blight are among the most damaging paddy diseases. Both are favoured by high humidity, extended leaf wetness and warm temperatures, which is why outbreaks cluster in humid monsoon periods. Bacterial leaf disease produces different lesion patterns and requires different management, so distinguishing the classes matters for treatment decisions.

2.2.2 Convolutional neural networks and transfer learning

CNNs learn hierarchical visual features directly from pixels. Residual networks such as ResNet-50 introduce skip connections that ease the training of deep architectures [3]. Transfer learning reuses weights learned on large image collections and fine-tunes them on a smaller target dataset, which is attractive when labelled field images are limited. Public repositories such as the PlantVillage dataset [2] have driven progress in plant-disease recognition, though their controlled backgrounds limit transfer to real fields.

2.2.3 Recurrent networks and LSTM

LSTM networks address the vanishing-gradient problem of simple recurrent networks by using gated memory cells that retain information over long sequences [6]. This makes them suitable for modelling how humidity, temperature and rainfall accumulate over days before a disease outbreak becomes visible.

2.2.4 Evaluation concepts

Classification quality is reported with precision, recall and F1-score, which are robust to class imbalance. Outbreak forecasting is a probabilistic binary problem evaluated with the area under the ROC curve (AUC) and calibration checks, so that predicted probabilities can be trusted when mapped to alert levels.

2.3 Related research

Lu et al. showed that deep CNNs can identify several rice diseases from leaf images with high accuracy under controlled conditions [1]. Hughes and Salathe released an open-access repository of plant-health images that became a standard benchmark for plant-disease classifiers [2]. Narmadha et al. studied rice blast detection using image-processing techniques [4], confirming the visual separability of blast lesions but within image-only pipelines. At the national level, remote-sensing studies of rice croplands [7] and the Department of Agriculture's paddy statistics [5] describe the extent and seasonality of production, which frames the scale at which disease losses matter.

Overall, the literature shows strong individual components: accurate CNN classifiers, effective sequence models for weather-driven risk, and established field observations that high humidity triggers blast. The synthesis in Table 2.1 summarises the approaches and their remaining gaps.

Table 2.1: Literature synthesis

Source / theme	Approach	Key finding	Gap
CNN rice disease classification [1], [4]	CNN / image processing	High accuracy on curated leaf images	Field image gap; image-only
Plant-health image repositories [2]	Open benchmark dataset	Enables transfer learning and comparison	Controlled backgrounds; non-local crops
Sequence modelling [6]	LSTM temporal modelling	Retains long-range dependence in sequences	Rarely applied to paddy disease with imagery
Mobile plant-disease apps	Commercial CNN apps	Good usability and scale	Mostly global datasets; post-symptomatic
Sri Lanka rice blast field observation	Field survey	High humidity is a trigger	No prediction model

2.4 Existing systems, products and techniques

Table 2.2 compares the approaches that farmers and researchers currently rely on. All of them detect disease after damage has begun or lack a link between environmental conditions and disease probability.

Table 2.2: Analysis of existing disease detection solutions

Solution	Data used	Strength	Key limitation
Plantix app	Mobile images	Good accuracy (about 92%)	Post-symptomatic only
Manual scouting	Visual inspection	Low cost	Late detection and inconsistent coverage
CNN research models	Laboratory images	High classification accuracy	Often not field-ready
Weather advisories	Climate data	Timely warnings	No direct disease correlation

2.5 Gap synthesis and conceptual framework

The reviewed work leaves three unresolved issues: image models are not robust to Sri Lankan field conditions, weather information is not converted into disease probability, and neither is delivered as an integrated, farmer-friendly service. The conceptual framework of this thesis closes the gap by coupling the two evidence streams: the CNN converts a leaf photograph into disease class and severity; these severity signals are added as features to the 14-day environmental sequence; and the LSTM converts the combined evidence into a seven- and fourteen-day outbreak probability, which decision rules translate into High, Medium or Low alerts. Visible symptoms and environmental triggers therefore reinforce each other, and the validation loop stores observed outbreaks and agronomist feedback for retraining and threshold tuning.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows an experimental design-science approach. The workflow iteratively builds and evaluates the CNN and LSTM models, refines alert thresholds using field feedback, and deploys the resulting services as interoperable modules within the smart agriculture platform. Quantitative evaluation (F1-score, AUC, latency) is complemented by qualitative farmer and extension-officer feedback collected during the pilot.

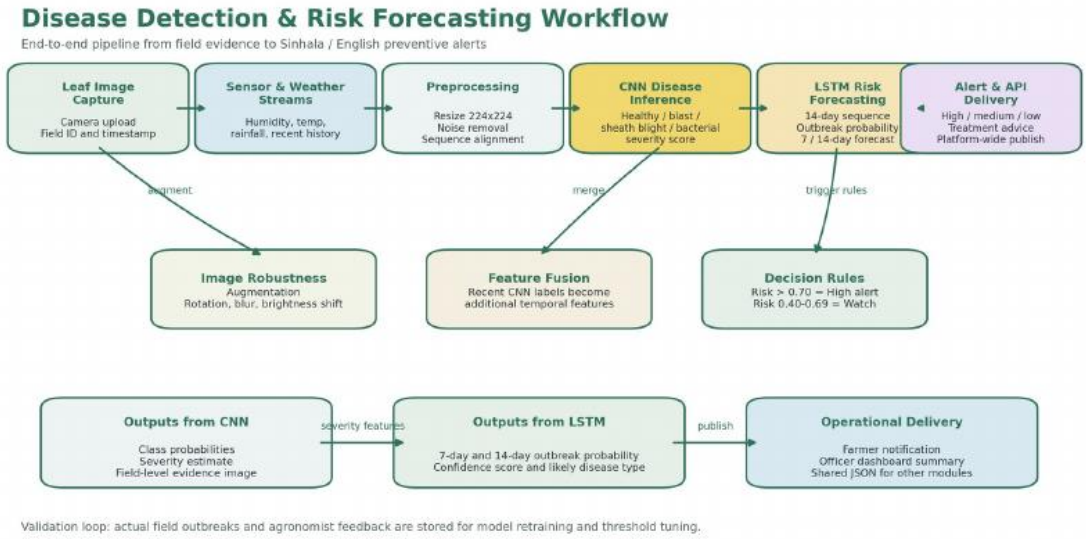


Figure 3.1: End-to-end workflow of the Disease Detection & Risk Forecasting component (image to CNN to LSTM risk to alerts)

Figure 3.1 summarises the pipeline: leaf image capture and sensor / weather streams are pre-processed, passed to CNN inference, fused with temporal features in the LSTM risk model, and converted into alerts and API outputs, with a validation loop feeding observed outbreaks back into retraining.

3.2 Data, participants and study materials

Image data. The target is at least 5,000 field images of healthy leaves and of blast, sheath blight and bacterial leaf disease, captured with smartphones and low-cost USB cameras in paddy fields. Each image carries a field ID and timestamp. Public images such as those in [2] may supplement training, but the held-out evaluation set consists only of field images from farms not seen during training.

Environmental data. Humidity, temperature and rainfall are collected from IoT sensor nodes and supplemented by Department of Meteorology feeds. Sensor records are time-stamped and aligned with image records and stored in PostgreSQL.

Labels. Image labels are assigned by the author and verified by Department of Agriculture agronomists or extension officers. Outbreak labels are derived from confirmed field observation within the 7- and 14-day horizons.

Participants. Paddy farmers and extension officers participate in the Yala 2026 pilot on the basis of informed consent. [Insert the number of farms, farmers, districts and sensor nodes actually involved.]

3.3 Experimental and study procedure

3.3.1 Image pipeline and CNN

Images are pre-processed with OpenCV: resizing to 224 x 224, normalisation, brightness adjustment, and targeted augmentation (rotation, blur and brightness shifts) to imitate field conditions. A ResNet-50 network pre-trained on a large image collection is fine-tuned for four-way classification of healthy, blast, sheath blight and bacterial leaf symptoms. The network also supplies a severity estimate and class probabilities.

3.3.2 LSTM risk forecasting

LSTM input sequences are built from 14 days of humidity, temperature and rainfall, plus the most recent CNN severity signals for the field. The output is the probability of an outbreak seven and fourteen days ahead, together with a confidence score and the likely disease type.

3.3.3 Ensemble logic and decision thresholds

CNN severity outputs are reused as LSTM features so that visible symptoms and environmental triggers reinforce each other. Alert thresholds are: risk above 0.70 triggers a High alert; risk between 0.40 and 0.69 is shown as Watch (medium) and surfaced for monitoring and officer review; lower values are Low. Thresholds are refined with field feedback.

3.3.4 Project schedule

Work is organised into five tasks (Table 3.1) aligned with the team schedule of Weeks 10-24 and the Yala 2026 season (Figure 3.2). Milestones are the CNN prototype by Week 12, LSTM integration and risk calibration by Week 16, field deployment from around Week 20, and the validation report at the end of the pilot.

Table 3.1: Work breakdown structure

Task	Description	Duration	Deliverable
1.0	Dataset collection	4 weeks	5,000 field images and synchronised environmental records
2.0	CNN development	5 weeks	Disease classifier with F1 >= 0.90 target
3.0	LSTM risk model	4 weeks	Risk model with AUC >= 0.85 target
4.0	API + app integration	3 weeks	Live Sinhala alerts and API endpoints
5.0	Field validation	6 weeks	Yala 2026 pilot report and user feedback summary

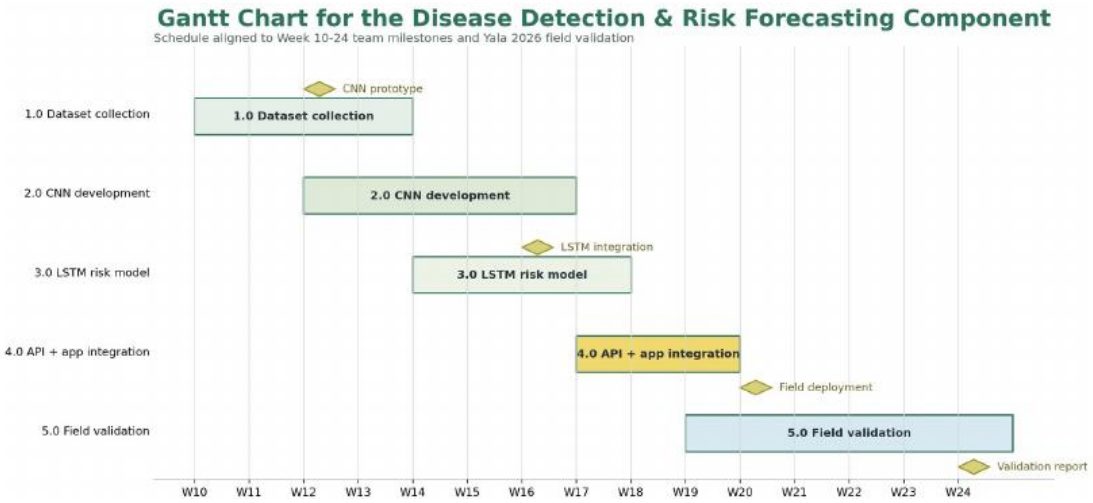


Figure 3.2: Gantt chart for the component schedule (Weeks 10-24)

3.4 Measures and evaluation metrics

Table 3.2 lists the measures used to evaluate each objective.

Table 3.2: Evaluation measures

Component	Metric	Target / method
CNN classifier	Precision, recall, F1-score, confusion matrix	F1 >= 0.90; five-fold cross-validation and held-out field set
LSTM forecaster	AUC, calibration, outbreak-probability (Brier) score	AUC >= 0.85; time-series splits
Fusion benefit	AUC difference vs weather-only LSTM	Paired comparison on same splits (H1)

Component	Metric	Target / method
System performance	Classification response time	Under 3 seconds per typical image
User validation	Farmer satisfaction and action logs	Above 80% satisfaction in pilot feedback

3.5 Data analysis

CNN performance is analysed with five-fold cross-validation and a held-out field test set covering unseen farms, reporting per-class precision, recall and F1-score and examining misclassifications by lighting, background and severity. LSTM evaluation uses chronological time-series splits to avoid leakage, calibration checks and outbreak-probability scoring. The fused model is compared against a weather-only LSTM baseline to test H1. Pilot alerts are compared against observed outbreaks and farmer action logs, and questionnaire responses are summarised descriptively. [Add statistical tests and confidence intervals used.]

3.6 Ethics, privacy, security and responsible AI

- Farmer participation is based on informed consent (see consent form in the appendices) and may be withdrawn at any time.
- Records are anonymised and uploaded imagery is access-controlled; location data is stored at field level only where needed.
- Data in transit is protected by HTTPS, and farmers are authenticated before accessing the service.
- Alerts display confidence levels and are advisory, with agronomist review for medium-risk cases, so that farmers are not asked to follow recommendations blindly.
- Datasets and model limitations (region, season, variety) are documented to avoid over-claiming generalisability.

[Add the ethics approval reference, if one was obtained.]

3.7 Reproducibility and research artefacts

Model code is developed in PyTorch 2.1 and OpenCV 4.8 with fixed random seeds, versioned datasets and documented hyper-parameters. Trained weights, preprocessing

scripts, the dataset schema, API specifications and the field-trial protocol are retained as research artefacts. [Insert repository / archive location.]

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

Table 4.1: Stakeholder identification

Stakeholder	Role	Need
Farmers	Primary users	Simple photo-to-alert workflow in local languages
Extension officers	Validation and advisory support	Dashboard risk summaries and farm prioritisation
DOA agronomists	Expert review and treatment guidance	Evidence-backed recommendations and feedback loop
Project team	Platform integration	Shared APIs and interoperable data objects

4.1.2 Functional requirements

- FR1: Accept leaf photos through the farmer mobile application camera or gallery upload.
- FR2: Classify disease type within approximately three seconds for a typical image submission.
- FR3: Generate seven- and fourteen-day outbreak risk forecasts using recent environmental conditions.
- FR4: Display Sinhala and English treatment alerts with confidence levels and recommended action windows.
- FR5: Share disease and risk outputs through platform APIs for use by linked modules and dashboards.

4.1.3 Non-functional requirements

Table 4.2: Non-functional requirements

Category	Requirement
Performance	Image classification response under 3 seconds for standard field submissions
Usability	Sinhala / English interface with offline-tolerant capture workflow
Reliability	High sensor-data capture rate and stable message synchronisation
Security	Encrypted farmer data and controlled authentication
Scalability	Ability to support multiple farms through service-based deployment

4.2 Component architecture within the complete solution

Figure 4.1 shows where the component sits in the platform. Farmer smartphones and IoT sensors feed the data and integration layer, which prepares features for the AI and analytics layer. A knowledge base of Department of Agriculture recommendations and localised treatment guidance supplies expert rules. Outputs travel to the delivery and user layer as Sinhala / English alerts, extension dashboards, district summaries and API outputs.

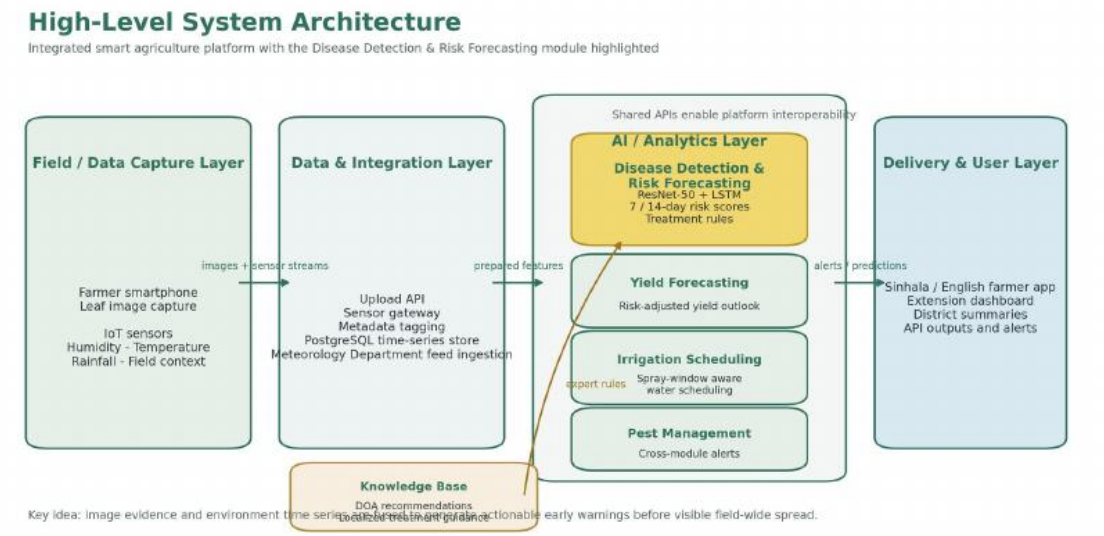


Figure 4.1: Platform architecture with the Disease Detection & Risk Forecasting module highlighted

Figure 4.2 details how the component exchanges data with other parties. Sensor data and farmer photos flow in; alerts return to the farmer app; risk summaries and expert feedback pass to and from extension / DOA review; and severity, risk and spray-timing information is shared with the yield forecasting and irrigation modules through shared APIs, without duplicating data capture.

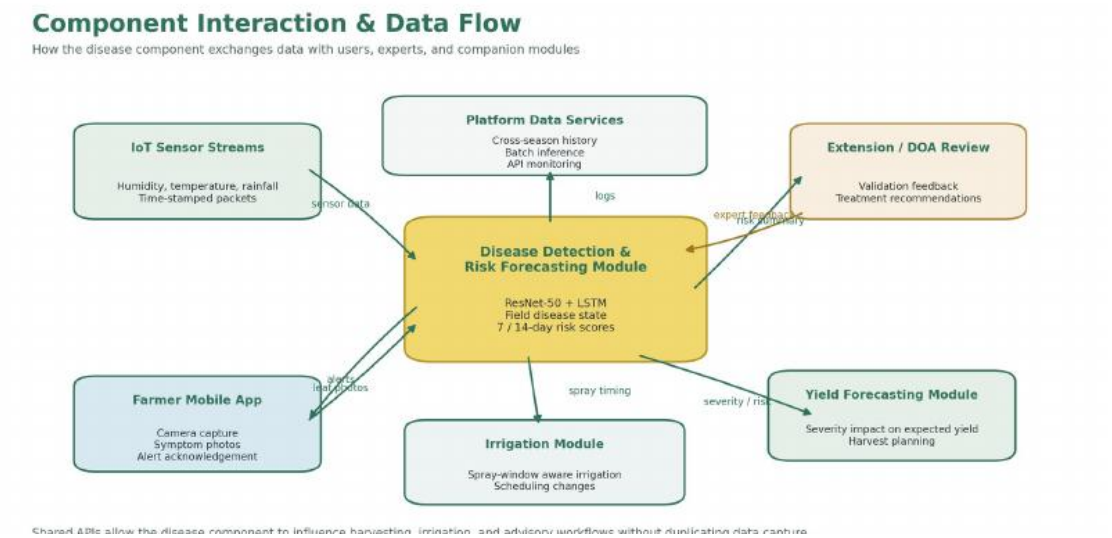


Figure 4.2: Component interaction and data flow among sensors, farmer app, expert review and companion modules

4.3 Design alternatives and decisions

Table 4.3: Design decisions and alternatives considered

Decision	Alternatives	Choice and rationale
Image model	Custom small CNN; ResNet-50 transfer learning	ResNet-50 transfer learning: strong features with limited labelled field data
Temporal model	GRU; LSTM; weather-threshold rules	LSTM: handles 14-day dependence and learns non-linear triggers; rules retained only as decision thresholds
Fusion	Separate CNN and weather alerts; fused features	CNN severity reused as LSTM feature so evidence reinforces each other
Deployment	Cloud-only; edge + cloud	Edge buffering on Raspberry Pi for low connectivity, cloud for training and hosting
Storage	Flat files; PostgreSQL	PostgreSQL for reliable time-series and image-linked records
Interface	English only; Sinhala / English	Sinhala / English for smallholder usability

4.4 Detailed design

Image module. Receives an uploaded photo, applies the OpenCV pipeline and runs ResNet-50 inference to produce class probabilities, a severity estimate and field-level evidence imagery.

Risk module. Aligns the latest 14-day sensor and weather sequence with recent CNN severity outputs and runs the LSTM to give 7-day and 14-day outbreak probabilities, a confidence score and the likely disease type.

Decision rules. Risk above 0.70 yields a High alert, 0.40-0.69 yields Watch, otherwise Low. Treatment advice is retrieved from the Department of Agriculture knowledge base and shown in Sinhala and English.

Delivery and API. Results are published as a farmer notification, an officer dashboard summary and a shared JSON object consumed by the yield, irrigation and pest modules. Microservices are containerised with Docker and use batch inference for scale.

4.5 Standards and professional practice

- Secure communication with HTTPS endpoints, authenticated access and anonymised logs.
- Consistent JSON API contracts to support interoperability between platform components.
- Use of open-source frameworks (PyTorch, OpenCV, PostgreSQL) with documented versions.
- Evaluation reporting aligned with accepted machine-learning practice (held-out sets, cross-validation, calibration).
- Ethical conduct in field research: consent, anonymisation and agronomist review of advice.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1: Technology stack

Tool	Purpose	Justification
PyTorch 2.1	CNN and LSTM development	Flexible deep-learning framework for training and deployment
OpenCV 4.8	Image preprocessing	Improves robustness for field image variability
PostgreSQL	Time-series and metadata storage	Reliable structured storage for sensor and image-linked records
Raspberry Pi 4	Edge inference and local buffering	Low-latency operation in field deployments
Cloud GPU (AWS)	Model training and hosting	On-demand compute for CNN and LSTM training
Docker	Service packaging	Reproducible, scalable microservice deployment

5.2 Component implementation

[Describe the actual implementation: dataset sizes achieved, CNN hyper-parameters (optimiser, learning rate, epochs, batch size, augmentation settings), LSTM architecture (layers, hidden units, sequence length 14), training environment, and any deviations from the design.]

The implementation follows the design in Chapter 4: an image service performing OpenCV preprocessing and ResNet-50 inference; a risk service that assembles the 14-day sequence and runs the LSTM; a rules layer that converts probabilities into alert levels; and an API service that exposes results as JSON.

5.3 Integration with the other components

The component integrates with the rest of the platform through shared APIs. The yield forecasting module consumes disease severity and risk for risk-adjusted yield outlook; the irrigation module consumes spray timing for spray-window-aware water scheduling; and the pest-management module receives cross-module alerts. Sensor streams arrive through the sensor gateway and upload API of the data and integration layer. [Insert the final JSON schema and endpoint list.]

Table 5.2: Interfaces with other platform components

Consumer	Data provided	Purpose
Yield forecasting	Severity, 7/14-day risk	Adjust expected yield and harvest planning
Irrigation scheduling	Spray timing, risk window	Avoid irrigation conflicts around spraying
Pest management	Cross-module alerts	Coordinate treatment advice
Farmer app / dashboard	Class, confidence, alert level, advice	Present Sinhala / English alerts and district summaries

5.4 Verification and software testing

Verification combines model evaluation with software testing. Model checks include five-fold cross-validation, held-out field-set evaluation and time-series split validation. Software testing covers unit tests for preprocessing and rule logic, API contract tests for JSON outputs, performance tests of classification latency against the three-second requirement, and offline / low-connectivity tests of edge buffering. Table 5.3 outlines the test plan.

Table 5.3: Test plan

ID	Test	Pass criterion
T1	Image upload via camera and gallery (FR1)	Image accepted and linked to field ID and timestamp
T2	Classification latency (FR2)	Typical image classified in about 3 seconds or less
T3	Risk forecast generation (FR3)	7- and 14-day probabilities returned for valid sequences
T4	Alert display (FR4)	Sinhala and English alert with confidence and action window
T5	API output (FR5)	JSON output valid and consumable by linked modules
T6	Offline buffering	No data loss during connectivity interruption

[Insert the actual test results and defects found and fixed.]

5.5 Deployment and operation

Services are deployed as Docker microservices with batch inference for scale. Raspberry Pi 4 nodes at the field provide edge inference and local buffering, synchronising to the cloud when connectivity returns. During the Yala 2026 pilot, alerts are delivered through the farmer application and the officer dashboard, and observed outbreaks and farmer

actions are logged for retraining and threshold tuning. [Describe the final deployment configuration and pilot operating period.]

6 RESULTS AND VALIDATION

[Results to be inserted after evaluation. The tables below are structured against the objectives and success indicators; replace the "Achieved" cells with measured values and add the supporting figures (confusion matrix, ROC curve, calibration plot).]

6.1 Research results

Tables 6.1 and 6.2 report the CNN and LSTM results against the targets set in Chapter 1.

Table 6.1: CNN classification results

Class / metric	Target	Achieved	Remarks
Healthy (F1)	>= 0.90	[ ]	[ ]
Rice blast (F1)	>= 0.90	[ ]	[ ]
Sheath blight (F1)	>= 0.90	[ ]	[ ]
Bacterial leaf disease (F1)	>= 0.90	[ ]	[ ]
Overall (macro F1), held-out field set	>= 0.90	[ ]	[ ]

Table 6.2: LSTM outbreak forecasting results

Model	Horizon	AUC	Calibration
Weather-only LSTM (baseline)	7 days	[ ]	[ ]
Weather-only LSTM (baseline)	14 days	[ ]	[ ]
Fused CNN + LSTM (proposed)	7 days	[ ]	[ ]
Fused CNN + LSTM (proposed)	14 days	[ ]	[ ]

6.2 Product and technical results

Table 6.3: Technical results against requirements

Requirement	Target	Achieved	Met?
Classification response time (FR2)	Under 3 seconds	[ ]	[ ]
Dataset size (SO1)	5,000+ field images	[ ]	[ ]
Sinhala / English alerts live (SO4)	Live alerts and JSON API	[ ]	[ ]

6.3 User or stakeholder validation

Validation during the Yala 2026 pilot compares generated alerts with observed outbreaks and farmer action logs and gathers satisfaction feedback from farmers and extension officers. The target is farmer satisfaction above 80%. [Insert number of respondents, satisfaction percentage, key comments and alert accuracy against observed outbreaks.]

6.4 Integration results

[Report the outcome of integration tests with the yield, irrigation and pest modules: successful API exchanges, latency, and issues resolved.]

6.5 Summary of findings

Table 6.4: Achievement against specific objectives

ID	Success indicator	Result	Status
SO1	Dataset of 5,000+ field images and synchronised sensor records	[ ]	[ ]
SO2	F1-score >= 0.90 on held-out evaluation	[ ]	[ ]
SO3	AUC >= 0.85 for outbreak prediction	[ ]	[ ]
SO4	Live Sinhala / English alerts and JSON API outputs	[ ]	[ ]
SO5	Farmer satisfaction above 80%	[ ]	[ ]

7 DISCUSSION

7.1 Interpretation of findings

[Interpret the results: whether the targets were met, how the fused model compares with the weather-only baseline (H1), which classes were hardest to separate, and what the pilot revealed about alert usefulness.]

7.2 Comparison with literature baselines and existing solutions

Existing laboratory CNN studies [1], [4] report high accuracy on curated images, and Plantix reports about 92% accuracy but identifies disease only after symptoms appear. The present component is distinguished not by competing on classification accuracy alone but by adding a forward-looking forecast built on Sri Lanka-specific environmental data. [Compare the achieved F1 and AUC numerically against these baselines.]

7.3 Research contribution

The research contributes a fused image-and-environment forecasting approach for paddy disease in Sri Lankan conditions, in which CNN severity signals act as features of an LSTM risk model, together with localised alert thresholds and an evaluation protocol based on unseen-farm field testing.

7.4 Product contribution

The product contribution is a deployable disease-intelligence service: a Sinhala / English photo-to-alert workflow for smallholders, edge-buffered sensor ingestion, and shared APIs that let the yield, irrigation and pest components use disease risk without duplicating data capture.

7.5 Limitations and threats to validity

- Limited to four leaf classes; other diseases and mixed infections are not modelled.
- Single-season pilot (Yala 2026) in a limited number of locations restricts external validity across agro-ecological zones, varieties and seasons.
- Outbreak labels depend on the timeliness and accuracy of field observation and agronomist confirmation.

- Image quality, lighting and leaf angle in farmer-captured photos may degrade performance despite augmentation.
- Sensor failure or connectivity gaps can reduce forecast quality, although edge buffering mitigates data loss.
- Farmer satisfaction measures perceived usefulness and may be influenced by novelty.

7.6 Implications

Early, evidence-based warnings can allow targeted preventive spraying instead of calendar-based or reactive spraying, reducing chemical use and cost. Integration with the other components shows how disease information can improve irrigation timing and yield planning, and the approach can be extended to other crops and diseases.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The immediate target market is Sri Lankan paddy smallholders who need timely disease warnings before visible crop loss becomes widespread. The value proposition is a locally relevant service that reduces guesswork by combining field imagery, environmental risk forecasting and treatment guidance in a Sinhala and English interface. Secondary users are extension services, cooperatives and institutional users who require district-level summaries.

8.2 Market and competitive position

The module differentiates itself by focusing on Sri Lankan paddy disease patterns, integrating localised environmental triggers, and delivering outputs through a workflow designed for smallholder farmers rather than laboratory-only validation. Its strongest advantage is coupling detection with forward-looking risk forecasting instead of post-symptomatic identification alone, whereas global apps such as Plantix remain reactive and weather advisories lack disease correlation.

8.3 Business and sustainability model

- Freemium mobile application with basic image-based detection available at no cost.
- Premium seasonal subscription for risk forecasts, preventive treatment plans and escalation alerts.
- District-level dashboards for cooperatives, extension services or institutional users needing summary analytics.

Table 8.1 gives the illustrative cost picture for early deployment, and Table 8.2 gives the component-level pilot budget. The budget remains modest because the modelling stack relies on open-source software and hardware needs are limited to field capture and edge deployment, making phased implementation feasible through project funding, institutional grants or shared team infrastructure.

Table 8.1: Illustrative cost estimation for early deployment

Category	Item	Illustrative cost (LKR)
Cloud	GPU training and model hosting	150,000

Category	Item	Illustrative cost (LKR)
Hardware	Raspberry Pi pilot nodes	80,000
Field testing	Validation and travel activities	50,000
Localisation	Sinhala UI preparation and content support	10,000

Table 8.2: Component pilot budget and justification

Item	Quantity	Total (LKR)	Justification
Raspberry Pi 4	3 units	90,000	Edge inference nodes and field buffering
USB cameras	3 units	15,000	Field image capture during pilot activities
Cloud GPU (AWS)	3 months	120,000	CNN and LSTM training workload
Sinhala font / localisation support	1	10,000	User interface preparation and content delivery
Total		235,000	Component-level pilot budget

8.4 Scalability, intellectual property and compliance

Dockerised microservices and batch inference allow scaling to many farms without changing the mobile workflow, and shared APIs let downstream modules consume risk scores without duplicating image or sensor capture. Intellectual property resides in the trained models, the curated local dataset and the fusion and threshold logic, subject to institutional policy and licences of open-source components and public datasets. Compliance requires consent, anonymisation and secure storage of farmer data, and attention to applicable Sri Lankan data-protection requirements. [Confirm IP ownership and applicable regulations with the supervisor.]

8.5 Contribution to the Sustainable Development Goals

Table 8.3: Contribution to the SDGs

SDG	Contribution
SDG 2: Zero Hunger	Reduces avoidable paddy losses and supports food security and sustainable agriculture
SDG 1: No Poverty	Protects smallholder income by limiting yield loss and unnecessary input cost
SDG 12: Responsible Consumption and Production	Targeted preventive spraying reduces chemical overuse
SDG 13: Climate Action	Supports adaptation to climate-driven variability in humidity and rainfall

SDG	Contribution
SDG 9: Industry, Innovation and Infrastructure	Applies IoT and AI innovation to agricultural infrastructure

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis addressed the lack of early, locally relevant disease warnings for Sri Lankan paddy by designing and developing the Disease Detection & Risk Forecasting component of an IoT and AI smart agriculture platform. The component integrates a ResNet-50 leaf classifier with an LSTM environmental risk model, delivers Sinhala and English alerts, and exposes results to the yield, irrigation and pest modules through shared APIs. [Summarise the final findings against the targets once results are available.]

9.2 Achievement of objectives

[Briefly restate whether SO1-SO5 were achieved, referring to Table 6.4.]

9.3 Individual research and product contributions

The individual research contribution is the fused image-and-weather outbreak forecasting approach adapted to Sri Lankan paddy conditions, with an evaluation based on unseen-farm field tests. The individual product contribution is the working detection and forecasting service, its alert logic, and the API through which other platform components use disease information.

9.4 Future directions

- Extend the dataset across more seasons, districts and rice varieties to improve generalisation.
- Add further diseases and mixed-infection detection, and finer-grained severity estimation.
- Explore alternative temporal architectures (for example GRU or attention-based models) and uncertainty estimation.
- Incorporate satellite or remote-sensing data and spatial spread modelling at district level.
- Strengthen offline, on-device inference for low-connectivity areas.
- Develop a scalable commercial deployment with cooperatives and extension services.

REFERENCES

[1] Y. Lu, S. Yi, N. Zeng, Y. Liu and Y. Zhang, "Identification of rice diseases using deep convolutional neural networks," Neurocomputing, [confirm volume, pages and year; the proposal lists 2020].

[2] D. P. Hughes and M. Salathe, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," arXiv preprint, 2015.

[3] K. He, X. Zhang, S. Ren and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778.

[4] R. Narmadha et al., "Rice blast disease detection using image processing," ICTACT Journal, 2021. [Add full title details, volume and pages.]

[5] Department of Agriculture, Sri Lanka, "Paddy Statistics Report," 2025.

[6] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735-1780, 1997.

[7] P. S. Thenkabail et al., "Global rice croplands mapping," Remote Sensing, 2022. [Confirm authors, volume and pages.]

[A thesis normally cites many more sources; add further peer-reviewed references, especially on field-condition rice disease datasets, LSTM disease forecasting, and mobile advisory tools. Cite every reference in the text and list no uncited sources.]

GLOSSARY

Augmentation - Artificial modification of training images (rotation, blur, brightness) to improve robustness.

Bacterial leaf disease - Paddy disease caused by bacteria, producing lesions distinct from fungal diseases.

Calibration - Agreement between predicted probabilities and observed outcome frequencies.

Edge inference - Running a trained model on a device near the data source rather than in the cloud.

Held-out set - Data reserved for final testing and never used during training.

Rice blast - Fungal paddy disease favoured by high humidity and prolonged leaf wetness.

Severity - Estimated extent of leaf damage caused by a disease.

Sheath blight - Fungal paddy disease affecting leaf sheaths, favoured by warm, humid, dense canopies.

Transfer learning - Reusing a network pre-trained on a large dataset as the starting point for a new task.

Yala / Maha - The two main paddy cultivation seasons in Sri Lanka.

APPENDICES

Appendix A Research instruments or extended results

The following supplementary materials accompany the research and are cited from the main text:

- A.1 CNN model architecture (Chapter 3 and Chapter 5).
- A.2 Risk forecasting workflow (Chapter 3).
- A.3 Sinhala UI mockups (Chapter 4).
- A.4 Field trial protocol (Chapter 3 and Chapter 6).
- A.5 Farmer consent forms (Section 3.6).

[Insert each appendix item here, plus any extended results, additional test cases and questionnaires.]

Appendix B Academic integrity and AI use disclosure

[Student to review and confirm this disclosure against module policy before submission.]

Table B.1: AI tool use disclosure

Tool	Purpose	Verification performed	Sections affected
Claude (Anthropic)	Restructuring the approved project proposal into the thesis template and drafting initial text	[Describe your review of facts, figures and citations]	Chapters 1-9 (draft text), front matter
[Other tools]	[Purpose]	[Verification]	[Sections]